

7. (a) The table shows some tests carried out to identify the ions contained in salts **A**, **B** and **C** and the observations made.

Salt	Flame test	Silver nitrate test	Name of salt
A	lilac flame	yellow precipitate	
B		no change	calcium nitrate
C			barium bromide

Complete the table by giving the name of salt **A** and the observations expected for salts **B** and **C**.

[3]

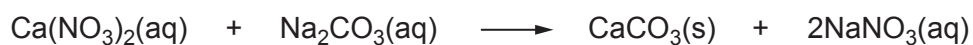
- (b) Give the numbers of protons and electrons in a calcium ion, Ca^{2+} , and a bromide ion, Br^- .

[2]

Ion	Number of protons	Number of electrons
Ca^{2+}		
Br^-		



- (c) Calcium nitrate solution reacts with sodium carbonate solution to form a precipitate of calcium carbonate.



Write an **ionic** equation for the formation of the calcium carbonate precipitate.

[2]

- (d) The solubility of barium bromide at different temperatures is shown in the table.

Temperature (°C)	Solubility (g / 100 cm ³ of water)
0	91
20	98
40	106
60	116
80	128
100	132

Calculate the mass of barium bromide that would crystallise when 500 cm³ of a saturated solution is cooled from 80 °C to 40 °C.

[2]

Mass = g

